

AMANGELDI, Kazakhstan

WMO: 353610

Lat: 50.133N

Lon: 65.233E

Elev: 466

StdP: 14.45

Time Zone: 6.00 (E06)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.7	-17.2	-28.6	1.1	-23.3	-22.8	1.6	-17.0	45.3	23.2	41.3	6.1	9.8	230	0.656

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.0	95.1	67.4	91.5	66.2	88.1	64.8	70.4	88.0	68.4	86.3	66.7	84.2	10.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	90.6	79.9	61.9	84.1	77.6	59.7	77.8	75.4	34.6	88.0	33.0	86.8	31.5	84.1	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 38.2	31.9	26.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.4	5.9	5.4	19.8	45.7	58.3	69.5	73.8	68.6	56.8	42.9	24.4	11.0
	DBStd	26.55	13.06	11.81	12.06	10.10	8.29	8.03	5.94	6.83	7.93	8.12	9.68	12.18
	HDD50	6026	1368	1248	937	197	23	0	0	0	26	249	769	1209
	HDD65	9655	1833	1668	1402	579	234	47	8	42	263	686	1219	1674
	CDD50	2508	0	0	0	68	281	585	738	577	231	28	0	0
	CDD65	661	0	0	0	0	27	181	281	154	18	0	0	0
	CDH74	8267	0	0	0	32	395	2405	3290	1854	284	7	0	0
	CDH80	3515	0	0	0	6	96	1097	1486	756	73	1	0	0
	WSAvg	10.7	13.5	12.7	13.2	11.3	10.2	9.3	9.0	9.0	9.0	9.5	10.8	10.7
	PrecAvg	17.1	1.5	1.4	1.5	1.2	1.3	1.3	1.9	1.2	0.8	1.3	1.7	1.8
(15) Precipitation	PrecMax	23.7	3.0	3.9	3.2	2.2	3.2	3.8	3.6	3.0	2.1	3.6	3.6	3.3
	PrecMin	12.2	0.4	0.2	0.1	0.1	0.3	0.3	0.6	0.1	0.0	0.2	0.6	0.7
	PrecStd	3.2	0.7	0.7	0.8	0.5	0.7	0.9	0.9	0.8	0.5	0.7	0.7	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	32.9	53.5	78.5	87.3	99.2	99.5	96.2	87.4	72.4	47.9	35.3
		MCWB	32.3	30.9	43.1	55.3	63.0	69.2	68.6	65.9	64.1	55.6	40.2	32.8
	2%	DB	30.5	29.9	44.4	72.3	82.8	95.0	95.3	91.3	81.5	66.3	42.0	32.4
		MCWB	29.4	28.5	38.2	54.4	63.3	67.9	67.6	64.2	59.2	51.8	38.7	30.7
	5%	DB	27.1	27.4	39.0	67.5	78.5	91.0	91.7	87.9	77.5	61.4	38.8	30.5
		MCWB	26.1	26.1	35.0	53.0	60.7	66.4	67.0	63.9	57.7	49.1	35.9	29.0
	10%	DB	24.0	24.6	35.2	62.7	74.7	87.0	88.2	84.3	73.2	57.0	36.0	28.1
		MCWB	22.9	23.4	32.4	50.6	58.3	64.6	65.4	62.4	56.1	46.7	33.8	26.7
	0.4%	WB	32.8	31.3	44.4	58.1	70.0	75.0	73.4	70.1	66.3	56.7	42.5	33.3
		MCDB	33.4	32.3	50.6	72.2	81.0	92.5	92.3	86.5	83.3	69.0	45.7	34.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	29.5	28.7	39.2	55.9	64.7	70.4	70.5	67.6	62.3	53.1	39.0	31.2
		MCDB	30.3	29.8	44.1	69.2	79.2	90.5	87.7	86.0	76.5	63.8	41.5	32.3
	5%	WB	26.1	26.1	35.3	53.7	61.8	67.8	68.7	65.4	59.2	50.1	36.1	29.2
		MCDB	27.0	27.3	38.8	65.5	75.1	86.7	86.2	83.2	73.5	60.0	38.6	30.6
	10%	WB	22.9	23.5	32.8	51.5	59.6	65.6	67.2	63.6	56.7	47.3	34.0	26.7
		MCDB	23.9	24.7	35.2	62.7	72.0	84.3	84.4	80.5	70.6	56.4	35.9	28.0
(35) Mean Daily Temperature Range	5% DB	MDBR	15.4	16.1	16.5	19.9	23.2	25.6	24.0	24.6	23.9	19.4	12.5	13.5
		MCDBR	16.5	15.6	17.6	27.6	30.1	30.6	29.1	30.7	31.0	28.1	14.6	14.4
		MCWBR	15.8	14.6	12.7	13.2	14.4	13.5	10.5	12.5	14.9	17.0	11.9	13.5
	5% WB	MCDBR	16.0	15.0	15.9	25.0	26.4	28.8	26.0	27.3	27.8	25.5	13.5	14.4
		MCWBR	15.5	14.4	12.1	13.5	14.0	14.3	11.7	12.6	14.8	16.4	11.5	13.6
(40) Clear-Sky Solar Irradiance	taub		0.306	0.332	0.393	0.423	0.417	0.415	0.420	0.413	0.362	0.350	0.340	0.310
	taud		2.051	2.000	2.003	2.096	2.189	2.256	2.257	2.255	2.346	2.295	2.184	2.089
	Ebn at Noon		218	244	250	259	267	268	264	260	262	241	206	196
	Edh at Noon		27	37	45	46	44	42	41	39	32	28	24	23
(44) All-Sky Solar Radiation	RadAvg		457	829	1214	1571	2029	2150	2017	1769	1341	763	366	302
	RadStd		54	66	128	138	118	121	127	130	83	85	39	44

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page